

KALINGAPATNAM, India

WMO: 431050

Lat: 18.333N

Lon: 84.133E

Elev: 6

StdP: 101.25

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	17.9	12.3	8.9	21.2	13.5	9.7	21.1	7.0	27.5	6.4	27.0	0.8	N/A	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.9	34.7	28.4	34.0	28.3	33.4	28.2	29.7	32.8	29.4	32.5	29.1	32.1	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.9	25.5	31.5	28.6	25.1	31.4	28.3	24.6	31.3	98.7	33.2	97.3	32.5	95.6	32.2	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.5	6.3	14.4	38.0	1.0	2.2	13.7	39.6	13.1	40.9	12.6	42.1	11.9	43.7		
			12.9	30.0	0.9	0.1	12.2	30.1	11.7	30.1	11.2	30.2	10.5	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	23.2	25.5	28.3	29.6	30.5	30.1	28.8	29.0	28.9	28.0	25.6	23.5
	DBStd	2.74	1.41	1.47	1.09	1.00	1.36	1.68	1.22	1.13	1.07	1.38	1.33	1.58
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6421	409	434	566	587	636	602	583	590	568	559	468	419
	CDD18.3	3379	151	201	308	337	378	352	325	332	318	301	218	161
	CDH23.3	38006	759	1495	3388	4367	5304	4757	3969	4069	3858	3352	1789	898
Wind	WSAvg	2.1	1.6	1.9	2.7	3.6	3.2	2.3	2.3	1.8	1.5	1.4	1.6	1.7
	PrecAvg	1557	11	15	22	47	108	196	311	314	274	187	58	16
	PrecMax	2177	50	71	106	91	359	352	409	474	565	529	197	117
	PrecMin	1212	0	0	3	9	31	88	177	207	157	40	1	0
	PrecStd	223	12	22	25	24	71	67	65	65	80	115	57	29
	DB	29.7	32.9	33.8	34.2	35.8	37.6	33.7	34.2	33.8	33.8	33.8	31.2	30.2
	MCWB	23.4	22.2	24.8	27.4	28.6	28.2	28.4	28.5	28.5	26.8	25.1	23.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.7	31.3	32.7	33.5	34.6	35.0	32.7	33.0	33.0	32.7	30.2	28.8	
	2% MCWB	22.8	23.6	25.7	27.6	28.7	29.1	28.2	28.3	28.3	26.7	24.6	23.1	
	5% DB	27.7	30.3	32.1	32.9	33.9	33.9	32.0	32.1	32.2	31.8	29.5	27.9	
	5% MCWB	22.3	23.7	26.0	27.5	28.8	28.8	28.0	28.0	28.0	26.6	24.3	22.6	
	10% DB	26.8	29.3	31.3	32.4	33.3	33.1	31.3	31.5	31.5	31.0	28.9	27.1	
	10% MCWB	22.0	23.5	25.9	27.4	28.7	28.6	27.8	27.9	27.8	26.5	24.0	22.2	
	WB	25.5	26.1	27.7	28.8	30.1	29.8	29.1	29.2	29.2	28.6	26.9	25.4	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	27.9	30.1	31.6	32.6	33.0	33.3	32.1	32.6	32.7	31.5	29.7	28.3	
	2% WB	24.1	25.4	27.2	28.4	29.7	29.6	28.7	28.7	28.7	28.0	26.2	24.7	
	2% MCDB	27.0	29.2	30.8	32.1	32.6	33.0	31.5	31.8	31.9	31.0	28.7	27.5	
	5% WB	23.3	24.9	26.8	28.1	29.5	29.3	28.3	28.3	28.3	27.6	25.7	23.9	
	5% MCDB	26.4	28.5	30.3	31.6	32.4	32.6	31.0	31.2	31.3	30.4	28.1	26.6	
	10% WB	22.7	24.3	26.5	27.9	29.2	29.0	28.0	28.1	28.0	27.2	25.1	23.2	
	10% MCDB	25.8	27.9	29.9	31.2	32.1	32.1	30.7	30.8	30.9	29.8	27.5	26.0	
Mean Daily Temperature Range	MDBR	8.5	9.0	7.1	6.2	5.9	5.3	4.4	4.8	5.1	5.9	7.0	7.9	
	5% DB MCDBR	9.2	10.1	7.9	6.7	6.4	6.5	5.7	6.0	6.3	7.2	7.8	8.4	
	5% DB MCWBR	4.5	4.2	2.9	2.6	2.8	2.8	2.6	2.7	2.8	2.7	3.5	3.8	
	5% WB MCDBR	7.8	8.2	6.6	5.9	5.3	6.0	5.1	5.5	5.8	5.9	6.1	6.9	
	5% WB MCWBR	4.4	3.9	2.8	2.6	2.7	2.9	2.6	2.8	2.9	2.8	3.3	3.9	
Clear-Sky Solar Irradiance	taub	0.526	0.521	0.599	0.600	0.648	0.659	0.580	0.554	0.554	0.550	0.536	0.552	
	taud	1.943	1.983	1.833	1.882	1.791	1.774	1.970	2.048	2.044	2.028	1.984	1.896	
	Ebn at Noon	751	785	738	741	698	683	741	766	764	751	737	710	
	Edh at Noon	177	179	214	206	223	224	185	172	171	169	169	181	
All-Sky Solar Radiation	RadAvg	4.65	5.55	6.10	6.59	6.09	4.58	4.15	4.26	4.51	4.68	4.48	4.25	
	RadStd	0.19	0.22	0.28	0.20	0.28	0.49	0.48	0.37	0.36	0.45	0.38	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.30	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page